EXPERIMENT NO. 9

Implement Polymorphism using Method Overloading and Overriding

PART B – TO BE COMPLETED BY STUDENTS

Roll No. C-60

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Class / Division: SE-C

Batch: C2

Date of Experiment _____

Date of Submission _____

Grade _____

B.1 Document Created by the Student

Add another specialized service and demonstrate dynamic method dispatch.

An additional specialized service named EmergencyAppointment is created by extending the Appointment class. Its service() method overrides the parent method. A parent-class reference is used to demonstrate dynamic method dispatch at runtime.

B.2 Student Program / Output

Paste/write the final program and attach or reproduce the observed output below.

```
class Appointment {
   void schedule(String patient) {
      System.out.println("Appointment for " + patient);
   }
   void schedule(String patient, String department) {
      System.out.println("Appointment for " + patient
            + " - " + department);
   }
   void service() {
```

```
        System.out.println("General appointment service");
    }
}
class TelemedicineAppointment extends Appointment {
    @Override
    void service() {
        System.out.println("Telemedicine appointment service");
    }
}
class EmergencyAppointment extends Appointment {
    @Override
    void service() {
        System.out.println("Emergency appointment service");
    }
}
public class PolymorphismDemo {
    public static void main(String[] args) {
        Appointment a = new Appointment();
        a.schedule("Meera");
        a.schedule("Meera", "Medicine");
        Appointment ref1 = new TelemedicineAppointment();
        Appointment ref2 = new EmergencyAppointment();
        ref1.service();
        ref2.service();
    }
}
```

```
Appointment for Meera
Appointment for Meera - Medicine
Telemedicine appointment service
Emergency appointment service
```

B.3 Observations and Learning

● Why is overloading called compile-time polymorphism?

Overloading is called compile-time polymorphism because the compiler determines which method to call based on the method arguments and parameter list.

● What is method overriding?

Method overriding occurs when a subclass provides its own implementation of an inherited method using the same method signature.

● What is dynamic method dispatch?

Dynamic method dispatch is the process in which Java selects the overridden method at runtime according to the actual object referenced by a parent-class variable**.**

Student observation / learning:

I learned that method overloading supports compile-time polymorphism, while method overriding supports runtime polymorphism. The program demonstrates dynamic method dispatch using telemedicine and emergency appointment services.

B.4 Conclusion

Telemedicine is an example of a healthcare application context mentioned by the course theme.

Student conclusion:

The experiment successfully demonstrates method overloading, method overriding, and dynamic method dispatch. Different specialized appointment objects execute their respective overridden methods through parent-class references.

B.5 Questions of Curiosity

6. Why is method overloading called compile-time polymorphism?

Because the compiler determines the appropriate overloaded method from its parameter list at compile time.

7. What is method overriding?

It is a feature in which a subclass provides a new implementation of an inherited method with the same signature.

8. What is dynamic method dispatch?

It is a runtime mechanism that selects the overridden method according to the actual object type.